

## 1. Executive Summary

|                         |        | Traffic Summary        |  | Count      |  |
|-------------------------|--------|------------------------|--|------------|--|
| Metric                  | Value  | Total Flows Observed   |  | 15822      |  |
| Total Threats Mitigated | 19     | Total Packets Analyzed |  | 2057523020 |  |
| High Priority           | 6      | Malicious Packets      |  | 2056517536 |  |
| Low Priority            | 13     | True Negatives Passed  |  | 11568      |  |
| Manual Releases         | 0      |                        |  |            |  |
| Manual Blocks           | 0      | Attack Vector          |  | Count      |  |
|                         |        | ICMP Flood             |  | 13         |  |
| ML Metrics              | Value  | SYN Flood              |  | 0          |  |
| False Positives         | 3      | UDP Flood              |  | 2          |  |
| FP Rate                 | 0.03%  | Uncertain              |  | 4          |  |
| IF Precision            | 99.93% |                        |  |            |  |
| IF Recall               | 96.32% | Action                 |  | Count      |  |
| IF F1-Score             | 98.09% | Quarantined            |  | 4          |  |
| IF Accuracy             | 98.96% | Time Ban               |  | 3          |  |
|                         |        | Blackhole              |  | 0          |  |
|                         |        | Released               |  | 2          |  |

## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value  | Description                                          |
|---------------------------|--------|------------------------------------------------------|
| Precision                 | 99.93% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 96.32% | Share of actual attacks successfully flagged         |
| F1-Score                  | 98.09% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 98.96% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.03%  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 3.68%  | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 96.32% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 99.97% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 4241           | FN 162            |
| Actual: Normal | FP 3              | TN 11406          |

### 2b. Random Forest: Attack Classification

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 99.95%  | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 100.00% | Share of actual attacks successfully flagged         |
| F1-Score                  | 99.98%  | Balanced measure combining Precision and Recall      |
| Accuracy                  | 99.95%  | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 66.67%  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 0.00%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 100.00% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 33.33%  | Normal traffic correctly identified as safe          |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 0              | 0               | 0              |
| Act: ICMP | 0              | 4241            | 0              |
| Act: UDP  | 0              | 0               | 0              |

### 2c. Hybrid Cascaded

| Metric                    | Value         | Description                                          |
|---------------------------|---------------|------------------------------------------------------|
| Precision                 | <b>99.93%</b> | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | <b>96.32%</b> | Share of actual attacks successfully flagged         |
| F1-Score                  | <b>98.09%</b> | Balanced measure combining Precision and Recall      |
| Accuracy                  | <b>98.96%</b> | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | <b>0.03%</b>  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | <b>3.68%</b>  | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | <b>96.32%</b> | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | <b>99.97%</b> | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | <b>TP 4241</b>    | <b>FN 162</b>     |
| Actual: Normal | <b>FP 3</b>       | <b>TN 11406</b>   |

### 2d. Response Latency

| Metric                   | Value           | Description                                                             |
|--------------------------|-----------------|-------------------------------------------------------------------------|
| Detection Time           | <b>507.0 ms</b> | Average time taken to detect an attack                                  |
| Mitigation Response Time | <b>5.5 ms</b>   | Interval between the anomaly flag and the FlowMod blocking rule install |

### 2e. Controller Resource Overhead

| Metric                         | Value        | Description                                                           |
|--------------------------------|--------------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | <b>1.12%</b> | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | <b>2.03%</b> | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | <b>1.35%</b> | Ryu controller CPU usage during simultaneous detection and mitigation |

## 3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.24 | 5        | 3            | 2026-09-26 23:03:59 | 2026-09-26 23:06:41 |
| 10.0.0.23 | 4        | 3            | 2026-09-26 23:03:58 | 2026-09-26 23:06:40 |
| 10.0.0.25 | 4        | 3            | 2026-09-26 23:04:01 | 2026-09-26 23:06:43 |
| 10.0.0.10 | 3        | 2            | 2026-09-26 23:00:49 | 2026-09-26 23:02:27 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector     | Confidence | Priority | Action         |
|---------------------|-----------|-------|------------|------------|----------|----------------|
| 2026-09-26 23:01:00 | 10.0.0.10 | DDoS  | Uncertain  | 68.4%      | Low      | Released       |
| 2026-09-26 23:01:25 | 10.0.0.10 | DDoS  | UDP Flood  | 85.9%      | High     | Time Ban (30s) |
| 2026-09-26 23:02:17 | 10.0.0.10 | DDoS  | UDP Flood  | 85.9%      | High     | Quarantined    |
| 2026-09-26 23:03:58 | 10.0.0.23 | DDoS  | Uncertain  | -          | Low      | Quarantined    |
| 2026-09-26 23:03:59 | 10.0.0.24 | DDoS  | Uncertain  | -          | Low      | Quarantined    |
| 2026-09-26 23:04:01 | 10.0.0.25 | DDoS  | Uncertain  | -          | Low      | Quarantined    |
| 2026-09-26 23:04:08 | 10.0.0.23 | DDoS  | ICMP Flood | 90.8%      | High     | Time Ban (30s) |
| 2026-09-26 23:04:08 | 10.0.0.23 | DDoS  | ICMP Flood | 90.8%      | Critical | Time Ban       |
| 2026-09-26 23:04:09 | 10.0.0.24 | DDoS  | ICMP Flood | 91.3%      | High     | Time Ban (30s) |
| 2026-09-26 23:04:10 | 10.0.0.24 | DDoS  | ICMP Flood | 91.3%      | Critical | Time Ban       |
| 2026-09-26 23:04:11 | 10.0.0.25 | DDoS  | ICMP Flood | 91.3%      | High     | Time Ban (30s) |
| 2026-09-26 23:04:12 | 10.0.0.25 | DDoS  | ICMP Flood | 91.3%      | Critical | Time Ban       |
| 2026-09-26 23:04:39 | 10.0.0.24 | DDoS  | ICMP Flood | 91.7%      | High     | Released       |
| 2026-09-26 23:04:50 | 10.0.0.24 | DDoS  | ICMP Flood | 91.7%      | Critical | Time Ban (1m)  |
| 2026-09-26 23:05:09 | 10.0.0.23 | DDoS  | ICMP Flood | 92.0%      | Critical | Time Ban (1m)  |
| 2026-09-26 23:05:12 | 10.0.0.25 | DDoS  | ICMP Flood | 92.0%      | Critical | Time Ban (1m)  |
| 2026-09-26 23:06:40 | 10.0.0.23 | DDoS  | ICMP Flood | 93.8%      | Critical | Time Ban (2m)  |
| 2026-09-26 23:06:41 | 10.0.0.24 | DDoS  | ICMP Flood | 91.2%      | Critical | Time Ban (5m)  |
| 2026-09-26 23:06:43 | 10.0.0.25 | DDoS  | ICMP Flood | 95.9%      | Critical | Time Ban (2m)  |

#### 5. IP Attack History (Completed Sessions)

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.25 | ICMP Flood | 0.6456   | 95.9% | Critical | Phase 2 | 1        | 11s      | 2026-09-26 23:06:43 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6461   | 91.2% | Critical | Phase 2 | 1        | 11s      | 2026-09-26 23:06:41 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6451   | 93.8% | Critical | Phase 2 | 1        | 11s      | 2026-09-26 23:06:40 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6127   | 92.6% | High     | Phase 2 | 1        | 1m 10s   | 2026-09-26 23:05:50 | Ban Expired                               |
| 10.0.0.25 | ICMP Flood | 0.6444   | 92.0% | Critical | Phase 2 | 3        | 1m 11s   | 2026-09-26 23:05:12 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6416   | 92.0% | Critical | Phase 2 | 3        | 1m 11s   | 2026-09-26 23:05:09 | Attack persisted - escalating to time ban |

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.24 | ICMP Flood | 0.6358   | 91.7% | Critical | Phase 2 | 3        | 10s      | 2026-09-26 23:04:50 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6027   | 91.4% | High     | Phase 2 | 2        | 41s      | 2026-09-26 23:04:42 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6072   | 91.7% | High     | Phase 2 | 1        | 40s      | 2026-09-26 23:04:39 | Ban Expired                               |
| 10.0.0.23 | ICMP Flood | 0.6447   | 90.8% | High     | Phase 2 | 2        | 40s      | 2026-09-26 23:04:38 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6456   | 91.3% | High     | Phase 2 | 1        | 10s      | 2026-09-26 23:04:11 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6456   | 91.3% | High     | Phase 2 | 1        | 10s      | 2026-09-26 23:04:09 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6451   | 90.8% | High     | Phase 2 | 1        | 10s      | 2026-09-26 23:04:08 | Attack persisted - escalating to time ban |
| 10.0.0.10 | UDP Flood  | 0.6220   | 85.9% | High     | Phase 1 | 2        | 10s      | 2026-09-26 23:02:27 | Attack Stopped                            |
| 10.0.0.10 | UDP Flood  | 0.5749   | 85.9% | High     | Phase 2 | 1        | 30s      | 2026-09-26 23:01:55 | Ban Expired                               |
| 10.0.0.10 | Uncertain  | 0.5811   | 68.4% | Low      | Phase 1 | 1        | 11s      | 2026-09-26 23:01:00 | Attack Stopped                            |

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE